

SOOT & STEEL

"Between Iron and Ice"



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Formal Game Proposal

1.1 Game Description

Name Soot & Steel

Tagline Between Iron and Ice

Genre Co-op management survival rogue-like

Perspective 2D top-down oblique projection

Nr. of players 4

Input method Xbox/DualSense Controller

1.1.1 Overview

Soot & Steel is a 4-player coop survival management game, where the players are tasked with moving a train with a wagon full of coal from one town to another in the cold Siberian tundra. On their journey, the players will encounter many perils, like hostile wildlife and bandits eyeing their cargo. To fend off against predators, the players may operate a set of train-mounted weaponry. Every part of the train must be operated manually. Accelerating and decelerating the train, fueling the coal-powered steam engine, reloading and aiming the turret, everything is achieved through manual button-prompts by a single player at a time. The cold of the tundra is a foe, too. Being shot by bandits may expose the interior to the cold arctic winds, which causes players to slowly freeze if the hull is not repaired. Damage to the engine can be fatal, as it's the main source of heat in the train. The core gameplay therefore involves the coordination of your crew mates and you, attempting to keep the train running and defend the cargo.

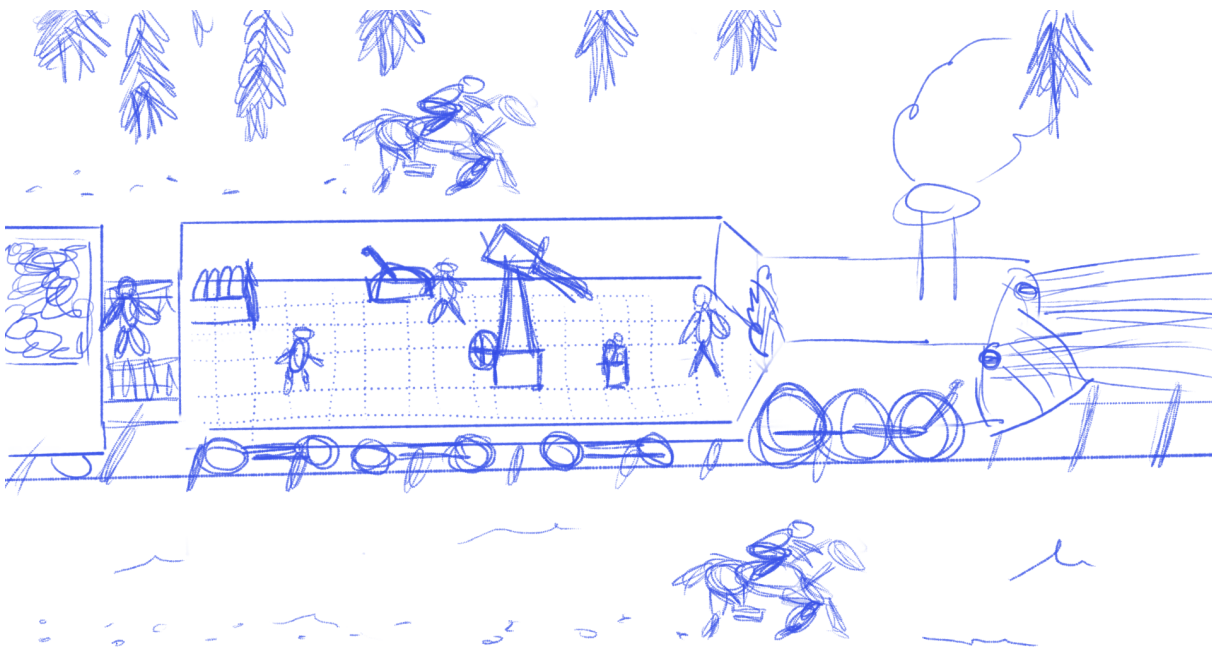
1 Formal Game Proposal

The auxiliary loop framing this combat-packed survival is management. With each stop, a part of the coal is used to recoup costs. Any profit can be used to upgrade the train or individual crew mates. Upgrades act to improve one aspect of the journey while complicating another. The players may choose to upgrade their weapon, with which foes can be taken care of more quickly, but may now need two people to operate. The players are incentivized to make smart choices within their budget and predict future perils.

1.1.2 Background Story

We're in post-revolution early Soviet Russia, ca. 1920s. The newly formed nation is embroiled in a civil war. The civilian population across the country is heavily dependent on coal, and as part of a Russian coal cooperative, it is your job to bring coal to the most outlying communities. However, the local forces, be they Bolshevik, Monarchist or local bandit groups, are looking for your coal to further their militaristic efforts. As this was mid-industrialization, they still mainly ride mounted on a horse, with bolt-action rifles. The towns are rural, far apart, and would surely not survive winter without your coal. In essence, your job is bitter, thankless, but crucial. The game, as such, should carry a sort of grittiness that reflects that reality, reflected in the title "Soot & Steel".

1.1.3 Design Decisions



For the combat portion of the game, anything on the train should be manual. To get the train going, coal from the wagon must be carried and shoveled into the engine, by pressing alternating buttons. Items can be carried on a button press, but the players speed is slowed when carrying. The speed of the train is set through a manual lever in the engine car, adjusted by pressing X or B. Depending on the speed, the engine burns more or less coal. Once outside of town limits, the cold sets in. Letting the engine die will cause players to start slowing down as the cabin becomes colder and colder, until all players freeze and lose the game. Throughout their journey, they may be attacked by one or more bandits trying to steal their coal. Those might come equipped with weaponry of all kind, like bolt-action rifles or even Molotov cocktails. The players must use their own weaponry, in the form of mounted turrets, to fend off the attackers. To turn the head of the turret, a wheel is placed at the base, which they may turn by circling with their joystick. To fire, they must pull on a lever by mashing a button, shooting more rounds the further they pull the lever. When the hull is damaged, players will also start slowing down, and a hull repair must be done by pressing a random sequence of buttons to weld a new piece of metal to the hull. Munitions, coal and metal are finite resources that the player must prepare before they set out on their journey.

The off-combat gameplay loop revolves around making choices for the next leg of the journey. Players are paid out currency based on the remaining coal in their wagon. They may then use their profits to stock up on resources, as well as purchase upgrades for their train, which may include static upgrades, like higher armor or more devastating munition, or even new equipment with new interactions, like different kinds of guns, easier controls for the train, easier means of transporting coal, etc. These pieces of equipment need to be placed physically in the train, for which the players have a limited grid of spaces. If development time permits, the players can choose the next leg of their journey, opting for a more perilous route that pays better or a less dangerous route with more modest reward.

The choice of a management-survival blend is to provide a natural way to pace the game. Highly stressful gameplay may become tiring, so we need to provide the player with short moments of respite. This also adds an additional dimension of strategy. As games like *PlateUp!* and *Overcooked* were huge inspirations, we are interested in making a game where the food orders of such games aren't imposed by the game but by emergent situations of the players' own choosing, i.e. "food order" follows naturally from the situation, rather than as an external constraint. Hence, the theme serves as a way to provide agency to the player while still imposing a set of actions that need to be done in time. Additionally, we believe that a major improvement that *PlateUp!* introduced compared to *Overcooked* is this persistent management of the space and the available equipment, allowing certain tasks to be automated, but introducing new tasks all the same. These "grey" decisions interest us as players and we want to carry that into our game, too.

1.1.4 Reference Games

Gameplay-wise, we look to *PlateUp!* and *Overcooked* for the combat portions, as we found their gameplay loop to be highly addictive and rewarding. Particularly we note how *Overcooked* will often give players more than one task to do, which forces players to communicate and triage. For the management portion of the game, we believe that games like *FTL: Faster Than*

1 Formal Game Proposal

Light, *Darkest Dungeon*, *Synthetik: Legion Rising* and *Slay the Spire* may serve as inspiration for how to design the overarching gameplay loop. Rewarding players for their performance with items that fundamentally change one aspect of the gameplay shakes things up and keeps it interesting.

In terms of style, we think games like *Papers, Please*, *Frostpunk* and *Metro Exodus*, with their dark, gritty and unforgiving atmosphere lend themselves great to what we are looking for.

Ultimately, what makes our game interesting is the unique blend between the fast-paced hectic coop coordination of *PlateUp!*, the rogue-like aspects and pacing of a game like *Slay the Spire* and the gritty atmosphere of *Frostpunk*.

1.2 "Big Idea" Bullseye

The primary conceptual idea revolves around the theme of this Siberian coal transport. So far, however, this is just another *FTL: Faster Than Light* but set on a train. However, by adding our innovation that everything must be manual, we add a new level of chaotic micromanagement to the game.



Figure 1.1: Big Idea Bullseye

1.3 Technical Achievement

We believe the upgrade system will provide a significant technological and, mainly, design challenge. The main aspect that draws people to play rogue-like games is the variety of unlockable power-ups. People are drawn to keep playing through the curiosity of finding new game-changing items. With our wish to make these physically placeable, as well as have an impact on gameplay, this becomes a difficult balancing act and could be difficult to implement technically in a serviceable fashion. Additionally, quantity and quality are both important. We want items that fundamentally change how the game is played, while also having enough of them to keep players coming back. However, if this is done right, the game's progression and replayability would certainly improve drastically.

1.4 Development Schedule

1.4.1 Layered Task Breakdown

Functional Minimum

We need 4-player movement, collisions, enemies and the interactable train. The train is not yet customizable, but the setup is enough to satisfy the combat game loop. We have enemies that spawn in and provide the core failure state. In this stage, the game is essentially an infinite train ride, where the player tries to survive as long as possible. Resources are infinite. Simple animated sprites and tilemaps need to be created, as well as one looping soundtrack and some sound effects.

With that, the game is playable, but hardly enticing.

Low Target

Now we add the management loop. We allow the train to stop, to be upgraded, and the game to be made up of multiple stages. This also introduces finite resources and the coal economy, i.e. selling coal for a profit to buy upgrades. The upgrades are entirely static, as is the interior of the train. They therefore have no impact on gameplay besides changing parameters like speed, damage, etc. We introduce different enemy types with different behaviors, like enemies that board the coal wagon or enemies that throw Molotov cocktails.

In this state, the core gameplay loop stands. We'd be OK with the result, but we would not have achieved the sort of addictiveness we were looking for.

Desired Target

We make the interior customizable. Equipment can be rearranged (save for the engine and the coal) and we introduce new items that change gameplay in more meaningful ways. The train car can be upgraded as a whole: think larger engine, more armor, larger coal wagon, etc. The aim is to have a variety as large as possible to give the player something to come back to.

Now, the game achieves what we set out to do and provides a, hopefully, rewarding gameplay loop to the player.

High Target

The more sauce we can add to the game, the better. We're thinking of a particle system, lighting effects, dynamic music that fades out when combat winds down, more traversable environments, character dialogue, menus that are physically placed in the world rather than pure UI, etc. The game should not only be complete but feel really nice to play. There should be noticeable visual and audio feedback for all player actions.

Extras

Here we would look to extend the game to be playable for 2-4 players. We also like the idea of being able to leave the train mid-combat, maybe for side missions along the route or to flip a train switch to move the train in a different direction. We would also look into networking, to allow the game to be played on separate machines over the internet. In this stage, we would also look into an overarching story, much how it's done in other rogue-like games. Allowing the train to be extended by multiple wagons would be very interesting. Separate game modes, like an infinite horde mode would add replayability as well.

1.4.2 Task List

Task	Who	Description	Hours
1	All	Brainstorm design	4
2	All	Write game proposal	3
3	All	Create physical prototype	6
4	All	4-player movement	3
5	Cashen, Ollie	Train scene mockup with collisions	8
6	Alex Staikov	Enemy behaviour	16
7	Cashen	Player/train interaction	16
8	Nicolas	Failure states & reset	16
9	Nicolas, Alex Staikov	Various game menus	20
10	Joshua	Functional target 2D assets	20
11	Ollie	Functional target sounds & music	12
12	Cashen	Static upgrades & economy	24
13	Alex Schlieper	Stages & difficulty progression	32
14	Alex Staikov	New enemy behaviour	20
15	Joshua	Low target 2D assets	32
16	Ollie	Low target sounds & music	16
17	Ollie, Nicolas	Management-loop menus & coordination between 4 players	32
18	Ollie, Cashen	Train customization menu	20
19	Alex Schlieper	Interactable weapons	16
20	Cashen	Interactable upgrades (not weapons)	16
21	Alex Schlieper	Holistic train/wagon upgrades	24
22	Alex Staikov	More in-depth enemy progression	30
23	Joshua	Desired target 2D assets	24
24	Ollie	Desired target sounds & music	24

Table 1.1: Task List

1.4.3 Timeline

Task	Who	Week														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	All	X	X													
2	All		X	X												
3	All			X												
4	All				X											
5	C/O				X	X										
6	ASta				X	X										
7	C					X										
8	N					X										
9	N/ASta					X	X									
10	J				X	X	X									
11	O					X	X									
12	C							X								
13	ASch							X	X							
14	ASta							X	X							
15	J							X	X	X						
16	O								X	X						
17	O/N								X	X						
18	O/C										X	X				
19	ASch										X	X				
20	C											X	X			
21	ASch												X	X		
22	ASta												X	X	X	
23	J										X	X	X	X	X	X
24	O										X	X	X	X	X	X

Table 1.2: Timeline

1.5 Assessment

The main strength of *Soot & Steel* will be its mix of hectic co-op coordination and meaningful management decisions. The coolest part should be the emergent chaos created when multiple systems fail at once and players must quickly decide whether to fuel the engine, repair the hull, fight off enemies, or protect their cargo. The game is aimed at players who enjoy cooperative games like *Overcooked* or *PlateUp!*, but who also want longer-term progression and strategy. In the game, the players operate the train manually, manage resources, defend against attacks, and make upgrade choices between stages. The world they find themselves in, should be a harsh, cold, and fragile industrial journey through the Siberian tundra, where survival depends on teamwork and constant trade-offs. We would judge the design a success if the manual interactions feel intuitive but stressful, the upgrade choices meaningfully affect play, and the game consistently creates tension, cooperation, and a strong gritty atmosphere.